Course: BTech **Semester:** 3

Prerequisite: Basic knowledge of Problem Solving,

Rationale : This course introduces Java programming and object-oriented concepts to build a strong foundation for developing reliable, platform-independent software applications and for pursuing advanced studies in software development.

Teaching and Examination Scheme

Teaching Scheme					Examination Scheme					Total
Lecture Hrs/Week	Tutorial Hrs/Week	Lab Hrs/Week	Seminar Hrs/Week	Credit	Internal Marks			External Marks		
					T	CE	P	T	P	
-	-	2	-	1	-	-	20	-	30	50

SEE - Semester End Examination, **T** - Theory, **P** - Practical

Course Outcome

After Learning the Course the students shall be able to:

1. Develop basic Java programs demonstrating program structure, variables, data types, type casting, operators, and control statements.
2. Implement iterative and recursive solutions using looping constructs, methods, and recursion to solve computational problems.
3. Design and manipulate one-dimensional and multi-dimensional arrays and perform array-based applications such as sorting, searching, and matrix operations.
4. Apply string handling techniques using String, StringBuffer, and StringBuilder classes and solve string-based problems.
5. Develop object-oriented Java programs using classes, objects, constructors, inheritance, polymorphism, abstraction, encapsulation, and keywords such as this and super.
6. Implement advanced Java features including exception handling, custom exceptions, collections framework, and multithreading for robust application development.

List of Practical	
1.	Write a Java program to display “Hello World” and demonstrate the Java program structure and demonstrate use of variables, data types, and type casting.
2.	Write a Java program to perform arithmetic, relational, and logical operations, bitwise and other operators.
3.	Write a Java program using below conditional statements. a) Even or Odd using if.. else statement. b) Roots of Quadratic Equation using else if ladder. c) Largest of three numbers using nested if else. d) Find out the week day using switch statement.
4.	Write a Java program to demonstrate looping constructs: a) Reverse of a number using while loop. b) Prime number using do while loop. c) nth term of fibonacci sequence using for loop.
5.	Write a Java program to perform methods and example on recursion.
6.	Write a Java program to perform operations on one-dimensional and multi-dimensional arrays.
7.	Write a Java program to demonstrate 1D - Arrays & 2-D Arrays. a) Maximum value and Second Maximum value without duplicates. b) Sort the names in Ascending Order. c) Addition of two matrix. d) 3x3 Matrix Multiplication.
8.	Write a Java program to demonstrate string handling using String, StringBuffer, and StringBuilder.
9.	Write a Java program to check the word is palindrome or not.
10.	Write a Java program to create a class and object and demonstrate constructors.
11.	Write a Java program to demonstrate encapsulation using access specifiers.
12.	Write a Java program to demonstrate inheritance: a) Single Inheritance. b) MultiLevel Inheritance. c) Hierarchial Inheritance. d) Hybrid Inheritance.
13.	Write a Java program to demonstrate use of the this and super keywords
14.	Write a Java program to demonstrate polymorphism (method overloading and overriding).
15.	Write a Java program to implement abstraction using abstract classes and interfaces(Multiple Inheritance).
16.	Write a Java program to demonstrate exception handling using try-catch-finally blocks.
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18.	Write a Java program to implement List interface using ArrayList and LinkedList.
19.	Write a Java program to implement Set interface using HashSet and TreeSet.
20.	Write a Java program to implement Queue interface using Priority Queue and Deque.
21.	Write a Java program to implement map interface using HashMap and LinkedHashMap.
22.	Write a Java program to demonstrate multithreading using Thread class and Runnable interface.